

Virtual Routing & Forwarding (Part-3)

- Topics covered:
 - Intro to VRF
 - VRF Configuration

Intro to VRF

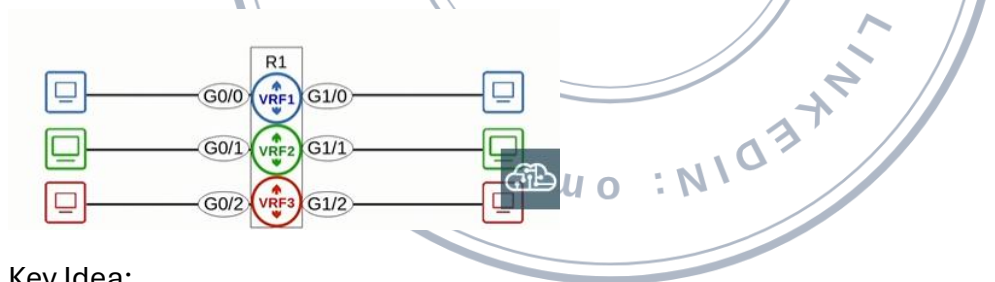
VRF (Virtual Routing and Forwarding)

This section introduces **VRF (Virtual Routing and Forwarding)** and its capabilities, followed by basic VRF configuration. Although VRF configuration is not a CCNA exam topic, seeing the configuration helps in understanding how VRF works.

VRF is not supported in Packet Tracer. To practice it, you need Cisco Modelling Labs (CML) or real Cisco devices.

Concept of VRF

The diagram below represents a single physical router (**R1**) divided into three virtual routers (VRF1, VRF2, and VRF3).



Key Idea:

- VRF divides a single physical router into multiple virtual routers.
- This is similar to how VLANs divide a single switch into multiple virtual LANs.

VLAN Comparison

Without VLANs:

- All switch interfaces belong to the same LAN.
- All devices share the same broadcast domain.

With VLANs:

- One broadcast domain is divided into multiple virtual broadcast domains.

Computer Networks - CCNA

Similarly, for routers:

- By default, all router interfaces belong to the same routing domain.
- Traffic received on one interface can be forwarded out of any other interface if the routing table permits it.

Routing Domain Example

Without VRF: Traffic entering G0/0 can be routed and forwarded out of G1/2.

With VRF:

- G0/0 belongs to VRF1
- G1/2 belongs to VRF3

Traffic cannot be forwarded between them.

How VRF Works

VRF allows a router to maintain *multiple routing tables*.

Without VRF: Router One Routing Table

With VRF: Router have VRF1/2/3 Routing Table

Important Points:

1. Layer 3 interfaces are assigned to specific VRFs.
2. A VRF is also called a **VRF Instance**.
3. VRF applies only to Layer 3 interfaces.
4. It does not apply to Layer 2 switch interfaces.
5. Router interfaces, SVIs, and routed ports on multilayer switches can be assigned to VRFs.

Traffic Between VRFs

Normally: VRF1 X VRF2 traffic cannot pass between different VRFs.

Exception: *VRF Leaking* can be configured to allow communication between VRFs.

This is an advanced topic and is not covered further here.

VRF-lite

VRF is commonly used with **MPLS (Multi-Protocol Label Switching)**.

However, this lesson focuses on: VRF-lite = VRF without MPLS

Whenever "VRF" is mentioned in this lesson, it refers to **VRF-lite**.

Computer Networks - CCNA

Why Service Providers Use VRF

VRF allows a service provider router to carry traffic for multiple customers.

Benefit 1: Traffic Isolation

Each customer connects to a separate virtual router.

Customer A --> [VRF1]

Customer B --> [VRF2]

Customer C --> [VRF3]

|
R1

Each customer's traffic remains isolated from the others.

Benefit 2: Overlapping IP Addresses

Different customers can use identical IP subnets.

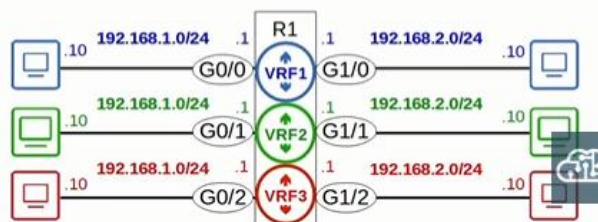
Customer Blue: 192.168.1.0/24, 192.168.2.0/24

Customer Green: 192.168.1.0/24, 192.168.2.0/24

Customer Red: 192.168.1.0/24, 192.168.2.0/24

With VRF, these overlapping subnets can coexist without problems.

Without VRF, this configuration is not possible on a router.

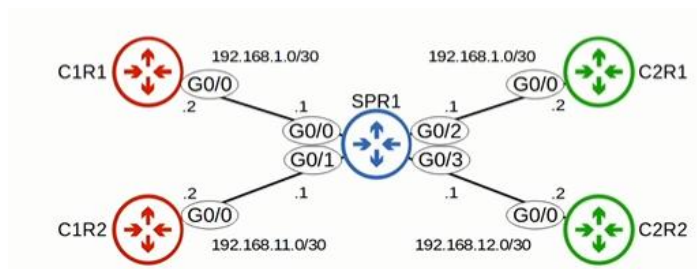


VRF Configuration

VRF Configuration Example

Network Topology

SPR1 provides WAN services for two customers.



Customer 1 uses 192.168.1.0/30

Customer 2 also uses 192.168.1.0/30

Attempt Without VRF

```
SPR1(config)# interface g0/0
SPR1(config-if)# ip address 192.168.1.1 255.255.255.252
SPR1(config-if)# no shutdown

SPR1(config-if)# interface g0/1
SPR1(config-if)# ip address 192.168.11.1 255.255.255.252
SPR1(config-if)# no shutdown

SPR1(config-if)# interface g0/2
SPR1(config-if)# ip address 192.168.1.1 255.255.255.252
% 192.168.1.0 overlaps with GigabitEthernet0/0

SPR1(config-if)# ip address 192.168.1.2 255.255.255.252
% 192.168.1.0 overlaps with GigabitEthernet0/0
```

G0/2 cannot use IP address 192.168.1.1 because it is in the same subnet as G0/0 (in this case it's the exact same IP address).

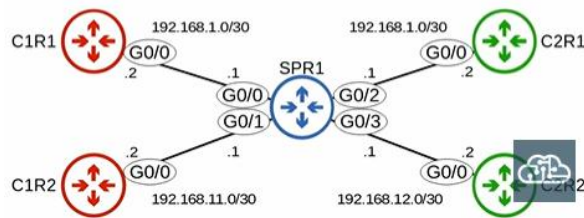
Even if the IP address is different, G0/2 cannot be configured in the same subnet as G0/0.

Explanation:

1. G0/0 and G0/1 are configured successfully.
2. Configuring G0/2 with 192.168.1.1/30 fails.
3. The subnet overlaps with G0/0.
4. Even changing the address to 192.168.1.2/30 fails.
5. Two interfaces on the same router cannot belong to the same subnet without VRF.

Computer Networks - CCNA

Creating VRFs



```
SPR1(config)# ip vrf CUSTOMER1
SPR1(config-vrf)# ip vrf CUSTOMER2
SPR1(config-vrf)# do show ip vrf
```

Name	Default RD	Interfaces
CUSTOMER1	<not set>	
CUSTOMER2	<not set>	

Explanation: VRFs are created in global configuration mode using: `ip vrf <VRF-NAME>`

Two VRFs are created:

1. CUSTOMER1
2. CUSTOMER2

`show ip vrf` confirms their creation.

Assigning Interfaces to VRFs (Configure G0/0)

```
SPR1(config-vrf)# interface g0/0
SPR1(config-if)# ip vrf forwarding CUSTOMER1
% Interface GigabitEthernet0/0 IPv4 disabled and address(es) removed due to enabling VRF CUSTOMER1
SPR1(config-if)# ip address 192.168.1.1 255.255.255.252
```

If an interface has an IP address configured, the

Explanation: When an interface is assigned to a VRF:

1. Any existing IP address is removed.
2. The IP address must be configured again afterward.

Configure G0/1

```
SPR1(config-if)# interface g0/1
SPR1(config-if)# ip vrf forwarding CUSTOMER1
% Interface GigabitEthernet0/1 IPv4 disabled and address(es) removed due to enabling VRF CUSTOMER1
SPR1(config-if)# ip address 192.168.11.1 255.255.255.252
```

If an interface has an IP address configured, the will be removed when you assign the interface to

Explanation: The previously configured IP address is removed and must be reconfigured.

Configure G0/2 and G0/3

```
SPR1(config-if)# interface g0/2
SPR1(config-if)# ip vrf forwarding CUSTOMER2
SPR1(config-if)# ip address 192.168.1.1 255.255.255.252
SPR1(config-if)# no shutdown
SPR1(config-if)# interface g0/3
SPR1(config-if)# ip vrf forwarding CUSTOMER2
SPR1(config-if)# ip address 192.168.12.1 255.255.255.252
SPR1(config-if)# no shutdown
```

Computer Networks - CCNA

Explanation: This time G0/2 successfully uses 192.168.1.1/30, even though G0/0 already uses the same subnet.

This works because: G0/0 -> CUSTOMER1 VRF and G0/2 -> CUSTOMER2 VRF

The interfaces belong to different routing tables.

Verify VRFs

```
SPR1(config-if)# do show ip vrf
Name           Default RD      Interfaces
CUSTOMER1      <not set>      Gi0/0
                <not set>      Gi0/1
CUSTOMER2      <not set>      Gi0/2
                <not set>      Gi0/3
```

Explanation: The output confirms: CUSTOMER1 VRF G0/0 and G0/1.

CUSTOMER2 VRF G0/2 and G0/3, SPR1 is now divided into two separate virtual routers.

Viewing VRF Routing Tables

After configuring the VRFs, show ip route was used on SPR1. Normally, after assigning IP addresses to interfaces, you would expect to see connected and local routes. However, no routes were displayed.

The reason is that show ip route displays the **global routing table**, which is the normal routing table used when VRFs are not configured.

Since all of SPR1's interfaces are assigned to VRFs, nothing appears in the global routing table.

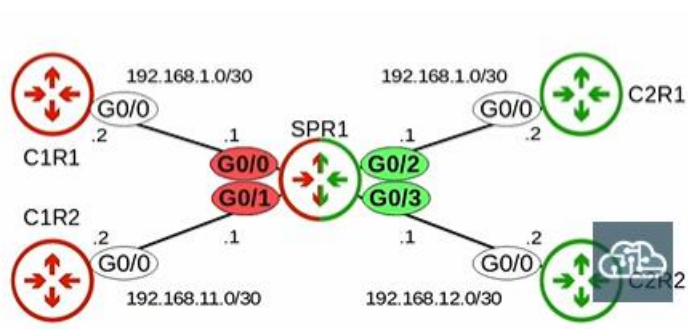
```
SPR1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set
```

show ip route displays the **global routing table**.
*All of SPR1's interfaces are configured in VRFs, so nothing displays here.
*You can have a mix of interfaces using and not using VRFs.

A router can have a mixture of interfaces that use VRFs and interfaces that do not.

Network Layout



Computer Networks - CCNA

Viewing CUSTOMER1 Routing Table

```
SPR1# show ip route vrf CUSTOMER1
Routing Table: CUSTOMER1
!output omitted

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/30 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.11.0/30 is directly connected, GigabitEthernet0/1
L    192.168.11.1/32 is directly connected, GigabitEthernet0/1
```

Explanation: Adding the VRF name to the command displays that VRF's routing table.

This output shows:

- Connected route for G0/0
- Local route for G0/0
- Connected route for G0/1
- Local route for G0/1

These routes belong only to the *CUSTOMER1 VRF*.

Viewing CUSTOMER2 Routing Table

```
SPR1# show ip route vrf CUSTOMER2
Routing Table: CUSTOMER2
!output omitted

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/30 is directly connected, GigabitEthernet0/2
L    192.168.1.1/32 is directly connected, GigabitEthernet0/2
192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/30 is directly connected, GigabitEthernet0/3
L    192.168.12.1/32 is directly connected, GigabitEthernet0/3
```

Explanation: This displays the routing table for the *CUSTOMER2 VRF*.

The routes shown belong only to:

1. G0/2
2. G0/3

These routes are separate from:

- The *CUSTOMER1* routing table
- The global routing table

Computer Networks - CCNA

Routing Table Separation

Global Routing Table: No Routes

CUSTOMER1 Routing Table:

- +-- 192.168.1.0/30 (G0/0)
- +-- 192.168.11.0/30 (G0/1)

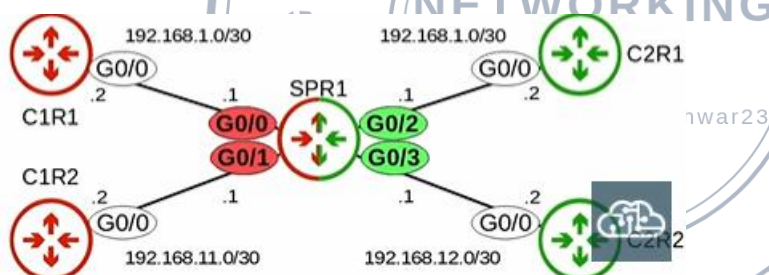
CUSTOMER2 Routing Table:

- +-- 192.168.1.0/30 (G0/2)
- +-- 192.168.12.0/30 (G0/3)

If an interface is not assigned to a VRF:

- Its routes appear in the global routing table.
- It remains isolated from interfaces in VRFs.
- Interfaces in different VRFs remain isolated from each other.

Testing Reachability



First Ping Attempt

```
SPR1# ping 192.168.1.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

Explanation: The ping failed because the global routing table contains no routes.

Without specifying a VRF, SPR1 uses the global routing table.

Computer Networks - CCNA

Ping Using CUSTOMER1 VRF

```
SPR1# ping vrf CUSTOMER1 192.168.1.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```



Explanation: The ping succeeds because the CUSTOMER1 routing table is used.

The destination reached is: C1R1 (192.168.1.2)

Although two devices use the address 192.168.1.2, the specified VRF determines which device is reached.

Ping C1R2

```
SPR1# ping vrf CUSTOMER1 192.168.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.11.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

Explanation: This successfully reaches, C1R2 (192.168.11.2), because it exists within the CUSTOMER1 VRF.

Ping C2R2 from CUSTOMER1 VRF

```
SPR1# ping vrf CUSTOMER1 192.168.12.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

Explanation: The ping fails.

The CUSTOMER1 routing table contains no route to 192.168.12.2.

That destination belongs to CUSTOMER2.

Ping Using CUSTOMER2 VRF

```
SPR1# ping vrf CUSTOMER2 192.168.1.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```



Explanation: This time the destination reached is: C2R1 (192.168.1.2)

The same IP address exists in both customer networks, but the VRF determines which one is used.

Ping C2R2

```
SPR1# ping vrf CUSTOMER2 192.168.12.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

Computer Networks - CCNA

Explanation: The ping succeeds because the destination exists in the CUSTOMER2 VRF and the correct routing table is being used.

Communication Summary

CUSTOMER1 VRF:

C1R1 <----> SPR1 <----> C1R2

Communication Works

CUSTOMER2 VRF:

C2R1 <----> SPR1 <----> C2R2

Communication Works

Between Different VRFs

CUSTOMER1 X CUSTOMER2

Communication does not work between different VRFs.

